

BALAZS HETENYI

Software Engineer | Building Technology That Makes a Difference

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PROFILE

Software Engineer with over five years of professional experience building secure, scalable software that supports research, operations and data-driven decision making. Currently working at the Wellcome Sanger Institute, developing systems that support world-leading genomics and cancer research.

Experienced across the full software development lifecycle, with expertise in Python, React, TypeScript, cloud technologies and API development. Passionate about solving meaningful problems through thoughtful design, collaboration and reliable engineering, with a focus on delivering maintainable solutions that create real-world impact.

WORK EXPERIENCE

COSMIC – Wellcome Sanger Institute, Cambridge, UK (Hybrid) – Nov 2022 – Present

Senior Software Developer – Nov 2024 – Present

Software Developer – Nov 2022 – Nov 2024

- **Promoted** due to constantly exceeding expectations across technical delivery and team contribution, including mentoring junior developers.
- **Led full-stack development of a document management system** for research data ingestion, owning architecture decisions across a FastAPI/PostgreSQL backend and React frontend, coordinating feature delivery end-to-end from database schema through to UI.
- **Built and maintained a research data management platform** using FastAPI, PostgreSQL, and Elasticsearch, implementing async background task pipelines with Celery/Redis for document ingestion, PubMed literature syncing, and S3 file management enabling researchers to ingest, search, and retrieve scientific documents through a unified, automated workflow.
- **Designed an Elasticsearch sync architecture** for full-text search across scientific documents, integrating two-phase S3 upload workflows with automatic indexing triggered via Celery task queues.
- **Delivered a production-grade async Python backend** with SQLAlchemy async sessions, Alembic migrations, and Docker-compose orchestration, supporting CI/CD deployment via GitLab.
- **Developed a genomic data platform** using Next.js/React and TypeScript, featuring an AWS S3-backed secure download system with pre-signed URLs, and Elasticsearch-powered curation workflows for cancer variant data empowering clinical curators to query and manage variants from the COSMIC database through a production web application.
- **Built full-stack curation tooling** for cancer integration and actionability workflows, integrating a Django/OAuth backend with JWT token refresh, custom API client abstraction, and multi-step form validation (Yup), enabling curators to upload, query, and manage clinical genomic data.
- **Delivered production-quality frontend infrastructure:** Storybook component library, Jest/React Testing Library tests, API client, auth flows, and GitLab CI with Husky pre-commit hooks on a large-scale bioinformatics web application serving cancer genomics data.

OpenClassrooms International, UK (Part-Time Remote) – Nov 2022 – Present

Web/Software Development Mentor

- **Mentored students** through full-stack projects in JavaScript, React, and Node.js, guiding them from fundamentals to production-ready application architecture.
- **Provided structured feedback** on frontend and backend design, API integration, and deployment, with an emphasis on scalability, clean code, and professional engineering practices.

Passle Ltd., Oxford, UK (Hybrid) – May 2021 – Nov 2022

Full-Stack Software Developer

- **Delivered customer-facing features** across a commercial SaaS platform using Angular, TypeScript, and C#.NET, contributing to a shared codebase across multiple product teams.
- **Built secure, reliable full-stack solutions** in close collaboration with product and design, developing strong foundations in agile delivery and cross-functional teamwork.

EDUCATION

BSc (Honours) Computing and IT (Software Engineering) – GPA: 82% (2:1)

The Open University – Oct 2020 – Dec 2025

Foundation Degree in Software Development

OpenClassrooms International – Sept 2018 – Sept 2019

SKILLS

Frontend Development: React, TypeScript, JavaScript, React Native, Next.js

Backend Development: Python, Django, FastAPI, C#, REST API

Databases: PostgreSQL (SQL Model/ SQLAlchemy), MongoDB

Cloud & DevOps: AWS, OpenStack, Docker, Celery, GitLab, CI/CD, Supabase

Testing & Quality: Cypress, React Testing Library, Jest, Pytest, Code Reviews

State management: Zustand

Collaboration: Agile / Scrum, Mentoring, Pair Programming, Stakeholder Communication

LANGUAGES

English – Fluent

Hungarian – Native

Spanish - Beginner